

Unit 1. Matrices, Linear Systems, and Column Relations

1. Matrices and column combinations

A real **matrix** $A = (a_{ij}) \in \mathbb{R}^{m \times n}$ has m rows and n columns; a_{ij} is its entry in row i , column j . Matrices are equal when their sizes and corresponding entries agree. A vector in $\mathbb{R}^n$ is a column:

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

For matrices of the same size and real c , define

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = ca_{ij}.$$

The zero matrix, denoted by 0 , has every entry zero; its size is determined by the equation in which it occurs. Set $-A = (-1)A$ and $A - B = A + (-B)$. These operations satisfy

$$\begin{aligned} A + B &= B + A, & (A + B) + C &= A + (B + C), \\ c(A + B) &= cA + cB, & (c + d)A &= cA + dA, \\ c(dA) &= (cd)A, & 1A &= A, \\ A + 0 &= A, & A + (-A) &= 0. \end{aligned}$$

Each identity holds entry by entry by the corresponding real-number law; for example, $c(a_{ij} + b_{ij}) = ca_{ij} + cb_{ij}$.

A **linear combination** of same-sized matrices $A_1, \dots, A_k$ is $c_1A_1 + \dots + c_kA_k$, where the coefficients are real. The empty combination is defined to be zero. A vector is a special case. Let $e_j \in \mathbb{R}^n$ have entry 1 in position j and zeros elsewhere. Comparing coordinate i gives

$$x = x_1e_1 + \dots + x_ne_n.$$

The product Ax

Write $A = [a_1 \ \dots \ a_n]$, with $a_j \in \mathbb{R}^m$. For $x \in \mathbb{R}^n$, define

$$Ax = x_1a_1 + \dots + x_na_n.$$

Its i th coordinate is $\sum_{j=1}^n a_{ij}x_j$. Consequently

$$Ax = b \iff \sum_{j=1}^n a_{ij}x_j = b_i \quad (1 \leq i \leq m).$$

This is a **linear system**: A is the coefficient matrix, $b \in \mathbb{R}^m$ is its right-hand side, and x is the unknown column. A solution must satisfy every equation. The system is **consistent** if a solution exists. For $b = 0$ it is **homogeneous**; then $x = 0$ is always a solution. A **nontrivial** solution is a nonzero column.

Example 1.1 (Representing a column). Let

$$A = \begin{pmatrix} 2 & -4 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ -2 \\ -3 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Determine whether $Ax = b$ and $Ax = c$ have solutions. The equality

$$2 \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} -4 \\ 2 \\ 0 \end{pmatrix} - 3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = b$$

shows that $Ax = b$ has a solution. For every x , coordinate 1 of Ax is minus twice coordinate 2:

$$2x_1 - 4x_2 = -2(-x_1 + 2x_2).$$

The column c fails this condition, since $1 \neq -2(0)$. Thus $Ax = c$ has no solution.

Proposition 1.1 (Linearity of multiplication by a matrix). For $A \in \mathbb{R}^{m \times n}$, $x, y \in \mathbb{R}^n$, and $s, t \in \mathbb{R}$,

$$A(sx + ty) = sAx + tAy.$$

Proof. By the column definition and the coordinate laws,

$$A(sx + ty) = \sum_j (sx_j + ty_j)a_j = s \sum_j x_j a_j + t \sum_j y_j a_j = sAx + tAy.$$

□

The same calculation gives $A \sum_{j=1}^k c_j x_j = \sum_{j=1}^k c_j Ax_j$ for any finite list: apply the two-term identity successively, with the empty case $A0 = 0$.

2. Matrix multiplication and transpose

For $A \in \mathbb{R}^{m \times n}$ and $B = [b_1 \ \cdots \ b_p] \in \mathbb{R}^{n \times p}$, define

$$AB = [Ab_1 \ \cdots \ Ab_p] \in \mathbb{R}^{m \times p}.$$

Column j of B supplies the coefficients for column j of AB . In entries,

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

The number of columns of A must equal the number of rows of B .

Example 1.2 (A rectangular product). For

$$A = \begin{pmatrix} 2 & -3 & 0 \\ 1 & 4 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 4 \\ -2 & 0 \\ 1 & 2 \end{pmatrix},$$

$$AB = \begin{pmatrix} 2(3) + (-3)(-2) + 0(1) & 2(4) + (-3)(0) + 0(2) \\ 1(3) + 4(-2) + 2(1) & 1(4) + 4(0) + 2(2) \end{pmatrix} = \begin{pmatrix} 12 & 8 \\ -3 & 8 \end{pmatrix}.$$

The two columns can also be obtained directly as combinations of the columns of A :

$$3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 2 \begin{pmatrix} -3 \\ 4 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ -3 \end{pmatrix}, \quad 4 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 8 \end{pmatrix}.$$

Proposition 1.2 (Product laws). For compatible matrix sizes and real c ,

$$\begin{aligned} A(B + C) &= AB + AC, & (A + B)C &= AC + BC, \\ (cA)B &= A(cB) = c(AB), & (AB)C &= A(BC). \end{aligned}$$

Proof. The first two identities follow from

$$\begin{aligned}\left(A(B+C)\right)_{ij} &= \sum_k a_{ik}(b_{kj} + c_{kj}) = \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj}, \\ \left((A+B)C\right)_{ij} &= \sum_k (a_{ik} + b_{ik})c_{kj} = \sum_k a_{ik}c_{kj} + \sum_k b_{ik}c_{kj}.\end{aligned}$$

For scalar compatibility,

$$\sum_k (ca_{ik})b_{kj} = c \sum_k a_{ik}b_{kj} = \sum_k a_{ik}(cb_{kj}).$$

For associativity take sizes $m \times n$, $n \times p$, $p \times q$. Both sides have size $m \times q$, and

$$\begin{aligned}\left((AB)C\right)_{ij} &= \sum_{k=1}^p \left(\sum_{\ell=1}^n a_{i\ell}b_{\ell k} \right) c_{kj} \\ &= \sum_{\ell=1}^n a_{i\ell} \left(\sum_{k=1}^p b_{\ell k}c_{kj} \right) = \left(A(BC)\right)_{ij}.\end{aligned}$$

The sums are finite, so their terms may be regrouped. Equality of all entries proves the claims. $\square$

In particular, $(AB)x = A(Bx)$: the factor nearest x acts first. The product of a matrix with a compatible zero matrix is zero, since each entry is a sum of zero terms.

Identity and powers

The **identity matrix** $I_n = [e_1 \cdots e_n]$ has ones on the diagonal and zeros elsewhere. The column definition gives $Ae_j = a_j$, so $AI_n = A$. Also, the (i, j) entry of $I_m A$ is a_{ij} , because only the i th term of its entry sum survives. Thus

$$I_m A = A = AI_n \quad (A \in \mathbb{R}^{m \times n}).$$

A matrix is **square** when its two sizes agree. For square A , set $A^0 = I$ and $A^{k+1} = A^k A$ for integers $k \geq 0$. Then

$$A^r A^s = A^{r+s} \quad (r, s \geq 0).$$

Indeed, for fixed r the case $s = 0$ is the identity law. If the formula holds for s , associativity gives $A^r A^{s+1} = A^r (A^s A) = (A^r A^s) A = A^{r+s+1}$.

Example 1.3 (Noncommutativity and failure of cancellation). Direct multiplication gives

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}.$$

Hence AB need not equal BA , even for square matrices of the same size. Such matrices **commute** when $AB = BA$.

For $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, multiplication gives $N^2 = 0$, although $N \neq 0$. Thus $NN = N0$ has unequal right factors: the common nonzero left factor N cannot be cancelled.

Transpose

The **transpose** of $A \in \mathbb{R}^{m \times n}$ is the matrix $A^\top \in \mathbb{R}^{n \times m}$ with $(A^\top)_{ij} = a_{ji}$. For example,

$$\begin{pmatrix} 2 & -3 & 0 \\ 1 & 4 & 2 \end{pmatrix}^\top = \begin{pmatrix} 2 & 1 \\ -3 & 4 \\ 0 & 2 \end{pmatrix}.$$

A square matrix is **symmetric** if $A^\top = A$. For columns $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$, the **outer product** $xy^\top$ is the $m \times n$ matrix with entries $x_i y_j$. For equal lengths, $x^\top y = \sum_i x_i y_i$; we identify a 1×1 matrix with its single entry.

Proposition 1.3 (Transpose laws). For compatible sizes,

$$(A^\top)^\top = A, \quad (A + B)^\top = A^\top + B^\top, \quad (cA)^\top = cA^\top, \quad (AB)^\top = B^\top A^\top.$$

Proof. The (i, j) entries in the first three identities are, respectively, a_{ij} , $a_{ji} + b_{ji}$, and ca_{ji} , on both sides. If A is $m \times n$ and B is $n \times p$, the last two matrices to compare have size $p \times m$, and

$$\left((AB)^\top\right)_{ij} = (AB)_{ji} = \sum_{k=1}^n a_{jk}b_{ki} = \sum_{k=1}^n (B^\top)_{ik}(A^\top)_{kj} = (B^\top A^\top)_{ij}.$$

□

3. Elimination and all solutions

The **augmented matrix** $[A \mid b]$ records a linear system. The three **elementary row operations** are

$$\begin{aligned} R_i &\leftrightarrow R_j && (i \neq j), \\ R_i &\leftarrow cR_i && (c \neq 0), \\ R_i &\leftarrow R_i + cR_j && (i \neq j). \end{aligned}$$

Each operation acts on an entire row, including the right-hand side. Two matrices are **row equivalent** if a finite sequence of these operations changes one into the other; the empty sequence is allowed.

Proposition 1.4 (Preservation of the solution set). An elementary row operation on $[A \mid b]$ preserves exactly the solutions of $Ax = b$.

Proof. Interchanging equations changes only their order. Multiplying an equation by nonzero c is undone by multiplying by $1/c$. For row replacement the old equations

$$\sum_k a_{ik}x_k = b_i, \quad \sum_k a_{jk}x_k = b_j$$

imply

$$\sum_k (a_{ik} + ca_{jk})x_k = b_i + cb_j, \quad \sum_k a_{jk}x_k = b_j.$$

Subtracting c times the unchanged second equation from the new first recovers the old first equation. Thus the implication holds in both directions for each operation, and hence for any finite sequence. □

Multiplication of a row by zero is excluded: it could erase an equation. In a replacement, the row being added remains unchanged.

Echelon form

A matrix is in **row echelon form** if all zero rows are at the bottom and the first nonzero entry of each successive nonzero row occurs strictly farther to the right. These first nonzero entries are the **pivots**; they need not be 1.

Theorem 1.5 (Finite elimination). Every real matrix is row equivalent to a matrix in row echelon form.

Proof. • If the matrix is zero, stop. Otherwise choose its leftmost nonzero column j . Swap rows to put a nonzero entry p in position $(1, j)$. All earlier columns are zero.

- For each $i > 1$, replace R_i by $R_i - (a_{ij}/p)R_1$. The new entry in column j is $a_{ij} - (a_{ij}/p)p = 0$; for $k > j$,

$$a'_{ik} = a_{ik} - \frac{a_{ij}}{p}a_{1k} \quad (k > j).$$

The resulting shape is

$$\begin{pmatrix} 0 & \cdots & 0 & p & * & \cdots & * \\ 0 & \cdots & 0 & 0 & & & \\ \vdots & & \vdots & \vdots & & B & \\ 0 & \cdots & 0 & 0 & & & \end{pmatrix}.$$

- Repeat within the lower-right matrix B . Operations between its rows preserve their earlier zeros, so each new pivot is strictly to the right of the preceding pivot. Each step fixes one row and removes at least one column from the remaining problem. After at most $\min(m, n)$ pivot steps, the remaining matrix is zero or has no rows or columns. The nonzero rows now have increasing pivot columns, and any remaining rows are zero: this is echelon form.

□

Example 1.4 (A skipped column and a redundant row). Reduce the following matrix, keeping the row operations visible:

$$\begin{aligned} \begin{pmatrix} 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -2 & 0 & 4 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix} &\xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix} \\ &\xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 & 3 \end{pmatrix} \\ &\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\ &\xrightarrow{R_1 \leftarrow R_1/2} \begin{pmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

The pivot columns are 1 and 3. Column 2 is not zero, but contains no pivot. A further replacement $R_1 \leftarrow R_1 + R_2$ clears the entry above the second pivot; this extra step is not required for echelon form.

For $Ax = b$, eliminate the coefficient columns and carry b along. Write the result as $[U \mid d]$, where U is in echelon form. Suppose its nonzero rows have pivot columns $j_1 < \cdots < j_r$. The unknowns x_{j_i} are the **pivot variables**; the remaining unknowns are **free variables**.

Theorem 1.6 (Consistency and free variables). The system $Ax = b$ is consistent if and only if every zero row of U has corresponding right-hand entry zero. If it is consistent, every choice of the $n - r$ free variables gives exactly one solution, and every solution is obtained this way. A consistent system has a unique solution exactly when there are no free variables.

Proof. • A zero coefficient row requires $0 = d_i$, proving necessity.

- Suppose all these conditions hold. Assign arbitrary real values to the free variables. For $i = r, r - 1, \dots, 1$, set

$$x_{j_i} = \frac{d_i - \sum_{k > j_i} u_{ik}x_k}{u_{ij_i}}.$$

The pivot u_{ij_i} is nonzero. Every variable on the right is free or has already been determined in a lower row. Each step therefore gives one value and satisfies row i ; the zero rows already hold. Reversibility gives a solution of $Ax = b$.

- Any solution has specific free-coordinate values. The same formulas force its pivot coordinates, so every solution is obtained. Different free-coordinate choices give different solutions. There is exactly one solution when there are no free variables; otherwise varying a free coordinate gives infinitely many.

□

Example 1.5 (All coefficients in a representation). For A, b in Example 1.1,

$$\left[\begin{array}{cccc|c} 2 & -4 & 0 & 0 & 4 \\ -1 & 2 & 0 & 0 & -2 \\ 0 & 0 & 1 & 2 & -3 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 + R_1/2} \left[\begin{array}{cccc|c} 2 & -4 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & -3 \end{array} \right].$$

Interchange rows 2 and 3. The nonzero equations are

$$2x_1 - 4x_2 = 4, \quad x_3 + 2x_4 = -3.$$

The free variables are $x_2 = s$, $x_4 = t$, giving all solutions

$$x = \begin{pmatrix} 2 + 2s \\ s \\ -3 - 2t \\ t \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -3 \\ 0 \end{pmatrix} + s \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ -2 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

Substitution gives $2(2 + 2s) - 4s = 4$, $-(2 + 2s) + 2s = -2$, and $(-3 - 2t) + 2t = -3$. Every solution has this form because its second and fourth coordinates are precisely s, t .

For right-hand side $c = (1, 0, 0)^\top$, the same first row operation instead produces a zero coefficient row with right-hand entry $1/2$. It requires $0 = 1/2$, proving inconsistency.

Proposition 1.7 (More unknowns than equations). If $A \in \mathbb{R}^{m \times n}$ and $n > m$, then $Ax = 0$ has a nonzero solution.

Proof. Row operations leave a zero right-hand side zero, so the homogeneous system is consistent. There are at most m pivots and hence fewer than n . Set one free variable to 1, the other free variables to 0, and obtain the pivot variables by back substitution. This gives a solution whose chosen free coordinate is nonzero. □

The condition is sufficient, not necessary. For example, $x_1 + x_2 = 0$, $2x_1 + 2x_2 = 0$ have solutions $(t, -t)^\top$ although there are two equations and two unknowns; substitution and the first equation prove that this is their entire solution set.

4. Column spaces, null spaces, and translated solutions

For $A \in \mathbb{R}^{m \times n}$, define the **column space** and **null space** by

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \quad N(A) = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n.$$

Membership $b \in \mathcal{C}(A)$ means that b is a combination of A 's columns, equivalently that $Ax = b$ is consistent. An element of $N(A)$ lists coefficients of a zero combination of those columns. The two sets generally lie in different coordinate spaces.

Proposition 1.8 (Closure under linear combinations). The sets $\mathcal{C}(A)$ and $N(A)$ contain zero and are closed under linear combinations.

Proof. The equality $A0 = 0$ puts zero in both sets. If $u = Ax$, $v = Ay$, then

$$su + tv = sAx + tAy = A(sx + ty) \in \mathcal{C}(A).$$

If $Ax = Ay = 0$, then $A(sx + ty) = sAx + tAy = 0$, so $sx + ty \in N(A)$. Repeating the two-term calculation proves closure for any finite combination; the empty one is zero. $\square$

Theorem 1.9 (The whole solution set). Suppose $Ax_0 = b$. Then

$$\{x : Ax = b\} = x_0 + N(A) := \{x_0 + z : z \in N(A)\}.$$

Proof. If $Ax = b$, then $A(x - x_0) = Ax - Ax_0 = b - b = 0$. Hence $x = x_0 + z$ with $z = x - x_0 \in N(A)$. Conversely, if $Az = 0$, then $A(x_0 + z) = Ax_0 + Az = b$. These two implications prove equality of the sets. $\square$

Example 1.6 (The sets for a rectangular matrix). For the matrix of Example 1.1,

$$\mathcal{C}(A) = \left\{ \begin{pmatrix} 2u \\ -u \\ v \end{pmatrix} : u, v \in \mathbb{R} \right\} = \{b \in \mathbb{R}^3 : b_1 = -2b_2\}.$$

Indeed $Ax = (2u, -u, v)^\top$ with $u = x_1 - 2x_2$, $v = x_3 + 2x_4$. Conversely every u, v is obtained by taking $x = (u, 0, v, 0)^\top$. The displayed equation on b is equivalent to this form by setting $u = -b_2$, $v = b_3$. Also,

$$N(A) = \left\{ s \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ -2 \\ 1 \end{pmatrix} : s, t \in \mathbb{R} \right\},$$

because $Ax = 0$ says $x_1 = 2x_2$, $x_3 = -2x_4$. Thus $\mathcal{C}(A) \subseteq \mathbb{R}^3$, whereas $N(A) \subseteq \mathbb{R}^4$. For $b = (4, -2, -3)^\top$, adding the particular solution $(2, 0, -3, 0)^\top$ to this null space reproduces all the coefficients in Example 1.5.

5. Independence and unique representation

A finite list $v_1, \dots, v_k$ of columns in the same $\mathbb{R}^m$ is **linearly independent** if

$$c_1v_1 + \dots + c_kv_k = 0 \implies c_1 = \dots = c_k = 0.$$

It is **linearly dependent** if a zero combination exists with at least one coefficient nonzero. The same definition applies to any list of matrices of the same size. The empty list is independent.

The standard columns $e_1, \dots, e_m$ are independent: coordinate i of $\sum_j c_j e_j = 0$ says $c_i = 0$. A list containing zero is dependent by giving that member coefficient 1 and all other members coefficient 0. A sublist of an independent list is independent: extend any relation in the sublist by zero coefficients on the omitted members and apply the original independence.

Proposition 1.10 (Dependence and a redundant member). A nonempty list is dependent if and only if some member is a linear combination of the others.

Proof. In a nonzero relation $\sum_i c_i v_i = 0$, choose j with $c_j \neq 0$. Solving for that member gives

$$v_j = -\sum_{i \neq j} \frac{c_i}{c_j} v_i.$$

Conversely, from $v_j = \sum_{i \neq j} d_i v_i$ obtain a zero combination with coefficient 1 on v_j and $-d_i$ on the others. For a one-member list the combination of the others is the empty sum, so both statements mean $v_1 = 0$. $\square$

Theorem 1.11 (Independence and uniqueness). Let $A = [a_1 \cdots a_n]$. The following conditions are equivalent:

- (a) The columns $a_1, \dots, a_n$ are linearly independent.
- (b) $N(A) = \{0\}$.
- (c) For every $b \in \mathcal{C}(A)$, the system $Ax = b$ has exactly one solution.

Proof. • (a) $\iff$ (b): $Ax = \sum_j x_j a_j$, so both conditions say that $Ax = 0$ forces $x = 0$.

- (b) $\implies$ (c): for $b \in \mathcal{C}(A)$, choose x_0 with $Ax_0 = b$. Theorem 1.9 gives the solution set $x_0 + N(A) = x_0 + \{0\} = \{x_0\}$.
- (c) $\implies$ (b): take $b = 0 \in \mathcal{C}(A)$. Since $A0 = 0$ and the solution is unique, $N(A) = \{0\}$.

□

The restriction $b \in \mathcal{C}(A)$ is essential. For $A = (1, 0)^\top$, the equation $Ax = 0$ forces $x = 0$, but $Ax = (0, 1)^\top$ is impossible by its second coordinate.

A list of more than m columns in $\mathbb{R}^m$ is dependent: put the columns into an $m \times n$ matrix with $n > m$ and apply Proposition 1.7 to obtain a nonzero coefficient relation.

Example 1.7 (Dependence at exceptional parameter values). For which real k are

$$v_1 = \begin{pmatrix} k \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ k \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ k \end{pmatrix}$$

independent? A relation $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$ says

$$k\alpha + \beta = 0, \quad \alpha + k\beta + \gamma = 0, \quad \beta + k\gamma = 0.$$

The first two equations give, without dividing by k ,

$$\beta = -k\alpha, \quad \gamma = (k^2 - 1)\alpha.$$

Substitution in the third gives

$$0 = -k\alpha + k(k^2 - 1)\alpha = k(k^2 - 2)\alpha.$$

If $k \notin \{0, \sqrt{2}, -\sqrt{2}\}$, this forces $\alpha = 0$, and then $\beta = \gamma = 0$. The vectors are independent. At each exceptional value, take $\alpha = 1$; the nonzero relation is

$$v_1 - kv_2 + (k^2 - 1)v_3 = 0.$$

Its three coordinates are $k - k = 0$, $1 - k^2 + k^2 - 1 = 0$, and $k(k^2 - 2) = 0$, respectively. Thus dependence occurs exactly for $k = 0, \pm\sqrt{2}$.

Exercises and solutions

Suggested practice: Exercises 1.3, 1.12, 1.15, 1.18, 1.21, and 1.24.

Matrix multiplication and transpose

Exercise 1.1. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 2 & 0 \\ 3 & -1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Compute $(AB)c$ and $A(Bc)$, showing the intermediate products.

Solution. The intermediate products have sizes 2×2 and 3×1 :

$$AB = \begin{pmatrix} 2+2-3 & 2+1 \\ 3+2+6 & 3-2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 11 & 1 \end{pmatrix}, \quad Bc = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}.$$

Consequently,

$$(AB)c = \begin{pmatrix} 1+9 \\ 11+3 \end{pmatrix} = \begin{pmatrix} 10 \\ 14 \end{pmatrix}, \quad A(Bc) = \begin{pmatrix} 8+2 \\ 12+2 \end{pmatrix} = \begin{pmatrix} 10 \\ 14 \end{pmatrix}.$$

Exercise 1.2. On an ordered triple (a_1, a_2, a_3) , let S reverse the order and let T move the last entry to the first position. Show that performing S then T need not give the same result as performing T then S . Do this first with triples, then with matrices acting on column vectors.

Solution. Applying S then T gives $(a_1, a_2, a_3) \mapsto (a_3, a_2, a_1) \mapsto (a_1, a_3, a_2)$. Applying T then S gives $(a_1, a_2, a_3) \mapsto (a_3, a_1, a_2) \mapsto (a_2, a_1, a_3)$. For instance, the triple $(1, 2, 3)$ gives different results. The corresponding matrices are

$$S = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The two orders give

$$TS = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad ST = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which are unequal.

Exercise 1.3. Construct real 2×2 matrices satisfying each condition:

- (a) $A^2 = -I$;
- (b) $B^2 = 0$ but $B \neq 0$;
- (c) $CD = -DC$ and $CD \neq 0$;
- (d) $EF = 0$, with every entry of both E and F nonzero.

Solution. (a) Take $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then $A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$.

(b) Take $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. It is nonzero, and direct multiplication gives $B^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

(c) Take

$$C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $CD = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $DC = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -CD$.

(d) Take $E = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $F = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Every entry is nonzero, but each entry of EF is either $1 - 1$ or $-1 + 1$, so $EF = 0$.

Exercise 1.4. Two square matrices A, B of the same size *commute* if $AB = BA$. Prove that

$$(A + B)(A - B) = A^2 - B^2$$

holds exactly when A and B commute.

Solution. Distributivity gives

$$(A + B)(A - B) = A^2 - AB + BA - B^2.$$

This equals $A^2 - B^2$ exactly when $-AB + BA = 0$, or $AB = BA$.

Exercise 1.5. (a) For symmetric $n \times n$ matrices A, B , prove that AB is symmetric exactly when $AB = BA$.

(b) Give two symmetric matrices whose product is not symmetric.

Solution. (a) Since $A^\top = A$ and $B^\top = B$,

$$(AB)^\top = B^\top A^\top = BA.$$

Thus $(AB)^\top = AB$ exactly when $BA = AB$.

(b) The symmetric matrices $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ give $AB = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$, which is not symmetric.

Exercise 1.6. A real square matrix K is *skew-symmetric* if $K^\top = -K$. Let A be a real square matrix.

(a) Prove that $A - A^\top$ is skew-symmetric.

(b) Express A as the sum of a symmetric matrix and a skew-symmetric matrix.

(c) Prove that every diagonal entry of a skew-symmetric matrix is zero.

Solution. (a) Entrywise transposition gives $(A - A^\top)^\top = A^\top - A = -(A - A^\top)$.

(b) Set

$$H = \frac{A + A^\top}{2}, \quad K = \frac{A - A^\top}{2}.$$

Then $A = H + K$, $H^\top = H$, and $K^\top = -K$.

(c) If $K^\top = -K$, the (i, i) entry gives $k_{ii} = -k_{ii}$. Hence $2k_{ii} = 0$, so $k_{ii} = 0$.

Exercise 1.7. For real columns a, b of the same length, determine exactly when $ab^\top = ba^\top$.

Solution. If $b = 0$, both matrices are zero. Suppose $b \neq 0$ and choose an index j with $b_j \neq 0$. Equality of the (i, j) entries gives $a_i b_j = b_i a_j$ for every i , so

$$a = \frac{a_j}{b_j} b.$$

Conversely, if $a = kb$ for a real k , then $ab^\top = kbb^\top = ba^\top$. The exact condition is therefore $b = 0$ or $a = kb$ for some real k ; this includes $a = 0$.

Exercise 1.8. Let $A = [a_1 \cdots a_n]$ be an $m \times n$ real matrix, with entries a_{rj} . Prove

$$AA^\top = \sum_{j=1}^n a_j a_j^\top, \quad (A^\top A)_{ij} = \sum_{r=1}^m a_{ri} a_{rj}.$$

Solution. The (r, s) entry of the first product is

$$(AA^\top)_{rs} = \sum_{j=1}^n a_{rj} a_{sj} = \sum_{j=1}^n (a_j a_j^\top)_{rs}.$$

For the second product, $(A^\top)_{ir} = a_{ri}$, so

$$(A^\top A)_{ij} = \sum_{r=1}^m (A^\top)_{ir} A_{rj} = \sum_{r=1}^m a_{ri} a_{rj}.$$

Both identities follow from the finite-sum definition of multiplication.

Exercise 1.9 (*). Let $A \in \mathbb{R}^{m \times n}$ and $B, C \in \mathbb{R}^{n \times p}$. Prove:

- (a) $A^\top A = 0$ if and only if $A = 0$;
- (b) $AB = 0$ if and only if $A^\top AB = 0$;
- (c) $AB = AC$ if and only if $A^\top AB = A^\top AC$.

Solution. (a) If $A^\top A = 0$, then, for every column j ,

$$0 = (A^\top A)_{jj} = \sum_{i=1}^m a_{ij}^2.$$

A sum of squares of real numbers is zero only when every summand is zero. Thus all entries of A vanish. The converse follows by multiplication.

(b) The forward implication follows by multiplying $AB = 0$ on the left by $A^\top$. Conversely, if $A^\top AB = 0$, then

$$(AB)^\top (AB) = B^\top A^\top AB = 0.$$

Part (a), applied to AB , gives $AB = 0$.

(c) Apply part (b) with $B - C$ in place of B :

$$AB = AC \iff A(B - C) = 0 \iff A^\top A(B - C) = 0 \iff A^\top AB = A^\top AC.$$

Exercise 1.10. Let $n \geq 1$, let $j = (1, \dots, 1)^\top \in \mathbb{R}^n$, and set

$$P = I_n - \frac{1}{n}jj^\top.$$

Prove that P is symmetric and *idempotent*, meaning $P^2 = P$. Explain these properties by describing Px for a column $x \in \mathbb{R}^n$.

Solution. Since $j^\top j = n$, associativity gives

$$(jj^\top)^2 = j(j^\top j)j^\top = njj^\top.$$

Therefore

$$P^2 = I_n - \frac{2}{n}jj^\top + \frac{1}{n^2}(jj^\top)^2 = I_n - \frac{1}{n}jj^\top = P.$$

Also $(jj^\top)^\top = jj^\top$, so $P^\top = P$. If $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, then

$$Px = x - \bar{x}j.$$

Thus P subtracts the mean from each coordinate. The result has coordinate sum $\sum_i (x_i - \bar{x}) = 0$, so applying P again changes nothing. Its diagonal entries are $1 - 1/n$ and its off-diagonal entries are $-1/n$: each coordinate contributes the same amount to every other coordinate, giving symmetry.

Solving systems and describing all solutions

Exercise 1.11. For each system, write its coefficient matrix, reduce it to row echelon form, and find all solutions.

(a) $2x - 3y + 4z = 0, 3x + y + z = 0.$

(b) $-x + 2y - 4z + w = 0, x + 3y + z - w = 0.$

Solution. (a) The coefficient matrix and a row reduction are

$$\begin{pmatrix} 2 & -3 & 4 \\ 3 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - \frac{3}{2}R_1} \begin{pmatrix} 2 & -3 & 4 \\ 0 & 11/2 & -5 \end{pmatrix}.$$

The second row gives $y = 10z/11$. Substituting into the first gives

$$2x = 3y - 4z = \frac{30}{11}z - 4z = -\frac{14}{11}z, \quad x = -\frac{7}{11}z.$$

Writing $z = 11t$, all solutions are $t(-7, 10, 11)^\top$, $t \in \mathbb{R}$.

(b) Here

$$\begin{pmatrix} -1 & 2 & -4 & 1 \\ 1 & 3 & 1 & -1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} -1 & 2 & -4 & 1 \\ 0 & 5 & -3 & 0 \end{pmatrix}.$$

Thus $y = 3z/5$, and the first row gives $x = 2y - 4z + w = 6z/5 - 4z + w = w - 14z/5$. With $z = 5s$ and $w = t$, all solutions are

$$(x, y, z, w)^\top = (-14s + t, 3s, 5s, t)^\top, \quad s, t \in \mathbb{R}.$$

In both parts the free coordinates can be chosen arbitrarily, and back substitution determines the remaining coordinates. Hence the displayed sets include every solution; substituting the formulas verifies them.

Exercise 1.12. Solve both systems by Gaussian elimination:

$$(a) \quad \begin{pmatrix} 2 & 1 & -3 \\ 5 & 2 & -6 \\ 3 & -1 & -4 \end{pmatrix} x = \begin{pmatrix} 1 \\ 5 \\ 7 \end{pmatrix}, \quad (b) \quad \begin{pmatrix} 2 & 1 & -2 \\ 3 & 2 & 2 \\ 5 & 4 & 3 \end{pmatrix} x = \begin{pmatrix} 10 \\ 1 \\ 4 \end{pmatrix}.$$

Solution. (a) Divide the first row of the augmented matrix by 2, then subtract 5 times the new first row from row two and 3 times it from row three:

$$\left(\begin{array}{ccc|c} 2 & 1 & -3 & 1 \\ 5 & 2 & -6 & 5 \\ 3 & -1 & -4 & 7 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 1/2 & -3/2 & 1/2 \\ 0 & -1/2 & 3/2 & 5/2 \\ 0 & -5/2 & 1/2 & 11/2 \end{array} \right).$$

Multiply row two by -2 , add $5/2$ times this row to row three, and divide row three by -7 . This gives

$$\left(\begin{array}{ccc|c} 1 & 1/2 & -3/2 & 1/2 \\ 0 & 1 & -3 & -5 \\ 0 & 0 & 1 & 1 \end{array} \right).$$

Back substitution yields $x_3 = 1$, $x_2 = -5 + 3 = -2$, and $x_1 = 1/2 - x_2/2 + 3x_3/2 = 3$.

(b) Divide row one by 2 and eliminate the entries below its first entry:

$$\left(\begin{array}{ccc|c} 2 & 1 & -2 & 10 \\ 3 & 2 & 2 & 1 \\ 5 & 4 & 3 & 4 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 1/2 & -1 & 5 \\ 0 & 1/2 & 5 & -14 \\ 0 & 3/2 & 8 & -21 \end{array} \right).$$

Multiply row two by 2, subtract $3/2$ times this row from row three, and divide row three by -7 :

$$\left(\begin{array}{ccc|c} 1 & 1/2 & -1 & 5 \\ 0 & 1 & 10 & -28 \\ 0 & 0 & 1 & -3 \end{array} \right).$$

Thus $x_3 = -3$, $x_2 = -28 - 10(-3) = 2$, and $x_1 = 5 - x_2/2 + x_3 = 1$.

Each echelon system determines all three coordinates, so the solutions are unique. Substitution into the original systems gives, respectively,

$$\begin{pmatrix} 6 - 2 - 3 \\ 15 - 4 - 6 \\ 9 + 2 - 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 7 \end{pmatrix}, \quad \begin{pmatrix} 2 + 2 + 6 \\ 3 + 4 - 6 \\ 5 + 8 - 9 \end{pmatrix} = \begin{pmatrix} 10 \\ 1 \\ 4 \end{pmatrix}.$$

Exercise 1.13. Let $A \in \mathbb{R}^{m \times n}$ and suppose $Ax = Ay = 0$. Prove that $x + y$ also solves the homogeneous system and that cx solves it for every real number c .

Solution. Distributivity and compatibility with scalar multiplication give

$$A(x + y) = Ax + Ay = 0 + 0 = 0, \quad A(cx) = c(Ax) = c0 = 0.$$

Thus both asserted columns belong to $N(A)$.

Exercise 1.14. Let $A \in \mathbb{R}^{m \times n}$ and suppose $Ax = b$. If $Ax' = b$ as well, prove that $x' = x + y$ for some y satisfying $Ay = 0$. Conversely, show that $x + y$ solves the original system whenever $Ay = 0$.

Solution. Set $y = x' - x$. Then $Ay = Ax' - Ax = b - b = 0$ and $x' = x + y$. Conversely, $A(x + y) = Ax + Ay = b + 0 = b$. Thus the full solution set is one particular solution plus all solutions of the homogeneous system.

Exercise 1.15. Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{m \times p}$, and suppose $AX_i = B$ for $i = 1, \dots, k$, where every X_i is $n \times p$. If real numbers $c_1, \dots, c_k$ satisfy $\sum_i c_i = 1$, prove that $\sum_i c_i X_i$ also solves $AX = B$.

Solution. All matrices in the sum have the same size. Distributivity gives

$$A \left(\sum_{i=1}^k c_i X_i \right) = \sum_{i=1}^k c_i A X_i = \sum_{i=1}^k c_i B = \left(\sum_{i=1}^k c_i \right) B = B.$$

Thus the asserted matrix is a solution, including when $B = 0$.

Consistency and exceptional parameters

Exercise 1.16. For which real numbers k does the system

$$x_1 + x_2 = 1, \quad 2x_1 + 2x_2 = k$$

have a solution? When it has a solution, is that solution unique?

Solution. Subtracting twice the first equation from the second gives $0 = k - 2$. Thus a solution exists exactly when $k = 2$. In that case the second equation repeats the first, and all solutions are $(1 - t, t)^\top$, $t \in \mathbb{R}$. There are infinitely many, so the solution is not unique.

Exercise 1.17. Can a matrix A be chosen so that $Ax = b$ is inconsistent for every right-hand side b of the appropriate size? Can a matrix A be chosen so that $Ax = b$ is consistent for every such b ? Justify each answer.

Solution. The first answer is no: for every matrix A , the right-hand side $b = 0$ has the solution $x = 0$. The second answer is yes: choose $A = I_m$. Then $x = b$ solves $Ax = b$ for every $b \in \mathbb{R}^m$.

Exercise 1.18. For each real k , find all solutions of the following systems:

- (a) $x_1 + 3x_2 = 0, \quad 2x_1 + kx_2 = 0;$
 (b) $x_1 + 2x_2 = 0, \quad 3x_1 + 6x_2 = 0, \quad 2x_1 + kx_2 = 0.$

Solution. (a) Subtracting twice the first equation from the second gives

$$x_1 + 3x_2 = 0, \quad (k - 6)x_2 = 0.$$

For $k \neq 6$, only $(x_1, x_2) = (0, 0)$ is possible. For $k = 6$, choose $x_2 = t$ freely; all solutions are $t(-3, 1)^\top$, $t \in \mathbb{R}$.

(b) Subtract three times the first equation from the second and twice the first from the third:

$$x_1 + 2x_2 = 0, \quad 0 = 0, \quad (k - 4)x_2 = 0.$$

For $k \neq 4$, the only solution is zero. For $k = 4$, all solutions are $t(-2, 1)^\top$, $t \in \mathbb{R}$. These reversible operations prove completeness; direct substitution checks each displayed family.

Exercise 1.19. Does the following system have a solution?

$$\begin{aligned}2x_1 + 3x_2 + x_3 &= 7, \\x_1 + 2x_2 - 2x_3 &= 8, \\3x_1 + 4x_2 + 4x_3 &= 20.\end{aligned}$$

Solution. The coefficient rows satisfy

$$2(2, 3, 1) - (1, 2, -2) = (3, 4, 4).$$

Thus twice the first left-hand side minus the second equals the third left-hand side. The same operation on the right-hand sides gives $2(7) - 8 = 6$, not 20. Subtracting twice row one and adding row two to row three therefore produces $0 = 14$. The system is inconsistent.

Exercise 1.20. For real parameters α, β , consider

$$\begin{aligned}\alpha x_1 + \beta x_2 + 2x_3 &= 1, \\ \alpha x_1 + (2\beta - 1)x_2 + 3x_3 &= 1, \\ \alpha x_1 + \beta x_2 + (\beta + 3)x_3 &= 2\beta - 1.\end{aligned}$$

When does a solution exist, and when is it unique?

Solution. Subtract the first equation from the second and third. The equivalent system is

$$\alpha x_1 + \beta x_2 + 2x_3 = 1, \quad (\beta - 1)x_2 + x_3 = 0, \quad (\beta + 1)x_3 = 2(\beta - 1).$$

- $\beta = -1$: the last equation is $0 = -4$, so no solution exists.
- $\beta = 1$: the equations reduce to $x_3 = 0$ and $\alpha x_1 + x_2 = 1$. For every α , all solutions are $(t, 1 - \alpha t, 0)^\top$, $t \in \mathbb{R}$.
- $\beta \neq \pm 1$: solve the last equation for x_3 , then the second for $x_2 = -x_3/(\beta - 1)$:

$$x_3 = \frac{2(\beta - 1)}{\beta + 1}, \quad x_2 = -\frac{2}{\beta + 1}.$$

Substituting into the first equation now gives

$$\alpha x_1 = 1 - \beta x_2 - 2x_3 = 1 + \frac{2\beta}{\beta + 1} - \frac{4(\beta - 1)}{\beta + 1} = \frac{5 - \beta}{\beta + 1}.$$

For $\alpha \neq 0$, this determines $x_1 = (5 - \beta)/(\alpha(\beta + 1))$, so the solution is unique. For $\alpha = 0$, it is consistent exactly when $\beta = 5$; then the solutions are $(t, -1/3, 4/3)^\top$, $t \in \mathbb{R}$.

Thus uniqueness holds exactly when $\alpha \neq 0$ and $\beta \neq \pm 1$. The remaining consistent cases are $\beta = 1$ with arbitrary α , and $(\alpha, \beta) = (0, 5)$; each has infinitely many solutions. All other parameter pairs give no solution.

Column sets and independence

Exercise 1.21. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad v = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

Prove the following descriptions, including both directions of each set equality:

- (a) $\mathcal{C}(A) = \{su + tv : s, t \in \mathbb{R}\}$ and $N(A^\top) = \{r(1, -2, 1)^\top : r \in \mathbb{R}\}$;
- (b) $\mathcal{C}(A^\top) = \mathbb{R}^2$ and $N(A) = \{0\}$;
- (c) $\mathcal{C}(AA^\top) = \mathcal{C}(A)$.

Solution. (a) For $y = (y_1, y_2)^\top$, put $s = y_1 + 2y_2$ and $t = 3y_1 + 4y_2$. Then

$$Ay = (s, t, 5y_1 + 6y_2)^\top = (s, t, -s + 2t)^\top = su + tv.$$

Conversely, any prescribed s, t arise from

$$y_1 = -2s + t, \quad y_2 = \frac{3s - t}{2}.$$

Substitution gives $y_1 + 2y_2 = s$ and $3y_1 + 4y_2 = t$, proving both inclusions in the first equality.

The equation $A^\top x = 0$ is

$$x_1 + 3x_2 + 5x_3 = 0, \quad 2x_1 + 4x_2 + 6x_3 = 0.$$

Subtracting twice the first equation from the second gives $-2x_2 - 4x_3 = 0$. Thus $x_2 = -2x_3$ and $x_1 = x_3$. Taking $x_3 = r$ gives exactly the asserted family.

(b) Given $(s, t)^\top \in \mathbb{R}^2$, choose

$$x = \left(\frac{3t}{2} - 2s, s - \frac{t}{2}, 0 \right)^\top.$$

Then $A^\top x = (s, t)^\top$, so every column in $\mathbb{R}^2$ lies in $\mathcal{C}(A^\top)$. The reverse inclusion follows from the size of $A^\top$.

If $Ay = 0$, its first two equations are $y_1 + 2y_2 = 0$ and $3y_1 + 4y_2 = 0$. Subtracting three times the first from the second gives $-2y_2 = 0$, and then $y_1 = 0$. Conversely, $A0 = 0$, so $N(A) = \{0\}$.

(c) For every $z \in \mathbb{R}^3$, $AA^\top z = A(A^\top z)$ belongs to $\mathcal{C}(A)$. This proves $\mathcal{C}(AA^\top) \subseteq \mathcal{C}(A)$. For the reverse inclusion, compute

$$AA^\top = \begin{pmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{pmatrix} = [d_1 \ d_2 \ d_3].$$

Its first two columns satisfy

$$\frac{25d_1 - 11d_2}{4} = u, \quad \frac{-11d_1 + 5d_2}{4} = v.$$

Consequently every $su + tv$ is a linear combination of d_1, d_2 . Part (a) now gives $\mathcal{C}(A) \subseteq \mathcal{C}(AA^\top)$.

Exercise 1.22. Let $a, b, c \in \mathbb{R}^m$ satisfy $a + b + c = 0$. Prove that

$$\mathcal{C}([a \ b]) = \mathcal{C}([a \ c]).$$

Solution. Since $b = -a - c$, every column in the first set has the form

$$sa + tb = sa + t(-a - c) = (s - t)a - tc,$$

and hence belongs to the second set. Conversely, $c = -a - b$, so

$$pa + qc = pa + q(-a - b) = (p - q)a - qb$$

belongs to the first set. The two inclusions prove the equality. Neither calculation requires any of the three columns to be nonzero or independent.

Exercise 1.23. Prove the following statements about finite lists of real column vectors:

- (a) The list consisting only of the zero column is linearly dependent.
- (b) Any list containing the zero column is linearly dependent.
- (c) A list of nonzero columns $x_1, \dots, x_n$ is linearly dependent if and only if one of its columns is a linear combination of the others.
- (d) A list of nonzero columns $x_1, \dots, x_n$ is linearly dependent if and only if some x_j , with $2 \leq j \leq n$, is a linear combination of $x_1, \dots, x_{j-1}$.

An empty sum is understood to be zero.

Solution. (a) The relation $1 \cdot 0 = 0$ has a nonzero coefficient.

(b) Give coefficient 1 to a zero column and coefficient 0 to every other column. This is a nontrivial relation.

(c) If $x_j = \sum_{i \neq j} a_i x_i$, subtract x_j to obtain

$$\sum_{i \neq j} a_i x_i - x_j = 0.$$

Its coefficient of x_j is -1 , so the list is dependent. Conversely, take a relation $\sum_i c_i x_i = 0$ with at least one $c_j \neq 0$. Solving for that column gives

$$x_j = - \sum_{i \neq j} \frac{c_i}{c_j} x_i.$$

(d) A relation $x_j = \sum_{i < j} a_i x_i$ gives a nontrivial relation on the full list by setting later coefficients to zero. Conversely, start with a nontrivial relation and let j be the largest index with $c_j \neq 0$. This index cannot be 1, since then $c_1 x_1 = 0$ would contradict $x_1 \neq 0$. All coefficients after j vanish, and therefore

$$x_j = - \sum_{i=1}^{j-1} \frac{c_i}{c_j} x_i.$$

For a one-column list with $x_1 \neq 0$, both sides of each equivalence in (c) and (d) are false, as required.

Exercise 1.24. (a) Prove that the columns of I_n are linearly independent.

(b) Show that the columns of

$$A = \begin{pmatrix} 1 & 3 & 2 & 0 \\ 1 & -5 & -3 & 1 \\ 5 & -1 & 0 & 2 \end{pmatrix}$$

are linearly dependent. Are its rows linearly dependent? For rows, dependence means a nontrivial linear combination equal to the zero row.

(c) A square matrix is *upper triangular* if its entries below the diagonal are zero, and *lower triangular* if its entries above the diagonal are zero. Prove that the columns of a triangular matrix are linearly independent exactly when every diagonal entry is nonzero.

Solution. (a) If $c_1e_1 + \cdots + c_ne_n = 0$, the i th coordinate is $c_i = 0$. Thus all coefficients vanish.

(b) Denote the columns by $a_1, \dots, a_4$ and the rows by r_1, r_2, r_3 . Direct calculation gives nontrivial relations:

$$3a_1 - a_2 - 8a_4 = \begin{pmatrix} 3 - 3 \\ 3 + 5 - 8 \\ 15 + 1 - 16 \end{pmatrix} = 0,$$

$$3r_1 + 2r_2 - r_3 = (3 + 2 - 5, 9 - 10 + 1, 6 - 6, 2 - 2) = 0.$$

Hence both lists are dependent.

(c) Let the triangular matrix be $T = (t_{ij})$. If all diagonal entries are nonzero, back substitution for an upper triangular matrix gives $x_n = 0, x_{n-1} = 0, \dots, x_1 = 0$ in $Tx = 0$. For a lower triangular matrix, solve from the first row downward instead. Thus only $x = 0$ solves the system, proving column independence.

For the converse, construct a nonzero solution whenever a diagonal entry is zero.

- Upper triangular: let k be the *first* zero diagonal index. Set $x_k = 1$ and $x_j = 0$ for $j > k$. Row k vanishes because $t_{kk} = 0$; rows below it involve only zero coordinates. For $i < k$, we have $t_{ii} \neq 0$, so define

$$x_i = -\frac{1}{t_{ii}} \sum_{j=i+1}^k t_{ij}x_j, \quad i = k-1, k-2, \dots, 1.$$

Each step makes row i vanish, and $x_k = 1$ ensures $x \neq 0$.

- Lower triangular: let k be the *last* zero diagonal index. Set $x_j = 0$ for $j < k$ and $x_k = 1$. Rows above k involve only zero coordinates; row k vanishes because $t_{kk} = 0$. All later diagonal entries are nonzero, so define

$$x_i = -\frac{1}{t_{ii}} \sum_{j=k}^{i-1} t_{ij}x_j, \quad i = k+1, k+2, \dots, n.$$

Again every row vanishes and $x_k = 1$.

Thus a zero diagonal entry gives a nonzero solution of $Tx = 0$, hence linearly dependent columns.

Exercise 1.25. Let A, B, C be linearly independent real $m \times n$ matrices. Determine whether $A + B$, $A + C$, and $B + C$ are linearly independent.

Solution. Suppose

$$\alpha(A + B) + \beta(A + C) + \gamma(B + C) = 0.$$

Collecting coefficients gives

$$(\alpha + \beta)A + (\alpha + \gamma)B + (\beta + \gamma)C = 0.$$

Independence of A, B, C implies

$$\alpha + \beta = 0, \quad \alpha + \gamma = 0, \quad \beta + \gamma = 0.$$

The first two equations give $\beta = \gamma = -\alpha$; the last then gives $-2\alpha = 0$. Hence $\alpha = \beta = \gamma = 0$. The three pairwise sums are therefore linearly independent.